

Mathematics for Computing

Module 2: Linear Algebra

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Topics Covered

1. Vectors
2. Coordinate System
3. Vector Addition
4. Vector Multiplication
5. Linear Combinations
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VECTORS

A vector is a mathematical object that has both:

- Magnitude
- Direction

A vector in $\mathbb{R}^2$ can be written as

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

A vector in $\mathbb{R}^3$:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

Vectors are fundamental representations of data in machine learning.

Vectors in AI and Machine Learning

A data point can be represented as a feature vector. For example, a house may be represented by

$$x = \begin{bmatrix} \text{area} \\ \text{bedrooms} \\ \text{age} \end{bmatrix}$$

For a numerical example:

$$x = \begin{bmatrix} 1500 \\ 3 \\ 10 \end{bmatrix}$$

A machine-learning dataset can therefore be represented as many vectors.

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{bmatrix}$$

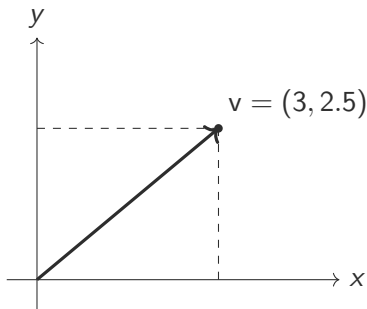
Coordinate System

COORDINATE SYSTEM

In a two-dimensional Cartesian coordinate system,

$$\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$$

represents a point (x, y) .



The coordinates specify the components of the vector relative to the chosen basis.

Magnitude of a Vector

For

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

the Euclidean magnitude is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}$$

For three dimensions:

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

Example

$$\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Then

$$\|\mathbf{v}\| = \sqrt{3^2 + 4^2} = 5$$

Vector Addition

VECTOR ADDITION Two vectors are added component by component. If

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

then

$$\boxed{\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}}$$

Example

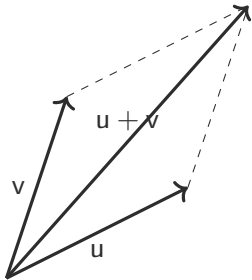
$$\mathbf{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

Therefore,

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

Geometric Interpretation of Vector Addition

Vector addition follows the parallelogram rule.



The diagonal represents the resultant vector.

$$u + v$$

Vector Multiplication by a Scalar

VECTOR MULTIPLICATION

Multiplication of a vector by a scalar changes its magnitude and possibly its direction.

$$c\mathbf{v} = \begin{bmatrix} cv_1 \\ cv_2 \end{bmatrix}$$

Example:

$$\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Then

$$3\mathbf{v} = \begin{bmatrix} 6 \\ 9 \end{bmatrix}$$

For $c < 0$, the direction is reversed.

Vector Linear Properties

For vectors u, v and scalars a, b :

$$a(u + v) = au + av$$

$$(a + b)u = au + bu$$

$$(ab)u = a(bu)$$

$$1u = u$$

These properties are the foundation of linear algebra.

Linear Combinations

LINEAR COMBINATIONS

A linear combination of vectors

$$v_1, v_2, \dots, v_k$$

is

$$a_1 v_1 + a_2 v_2 + \dots + a_k v_k$$

where $a_1, \dots, a_k$ are scalars.

Example

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Then

$$3v_1 + 4v_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Linear Combinations in Machine Learning

A linear model is itself a linear combination of feature vectors.

For features

$$x_1, x_2, \dots, x_d$$

a prediction may be

$$\hat{y} = w_1x_1 + w_2x_2 + \dots + w_dx_d + b$$

or

$$\hat{y} = \mathbf{w}^T \mathbf{x} + b$$

The weights determine how strongly each feature contributes.

This idea appears in:

- ▶ Linear regression
- ▶ Logistic regression
- ▶ Neural networks
- ▶ Support vector machines

SPAN AND BASIS The **span** of a set of vectors is the set of all possible linear combinations of those vectors. For vectors $v_1, \dots, v_k$:

$$\text{span}\{v_1, \dots, v_k\} = \left\{ \sum_{i=1}^k a_i v_i \right\}$$

Example:

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Then

$$ae_1 + be_2 = \begin{bmatrix} a \\ b \end{bmatrix}$$

Therefore,

$$\text{span}\{e_1, e_2\} = \mathbb{R}^2$$

Basis

A basis is a set of vectors that:

1. Spans the vector space.
2. Is linearly independent.

The standard basis of $\mathbb{R}^3$ is

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Every vector in $\mathbb{R}^3$ can be represented as

$$v = v_1 e_1 + v_2 e_2 + v_3 e_3$$

Linear Independence

Vectors $v_1, \dots, v_k$ are linearly independent if

$$a_1 v_1 + \dots + a_k v_k = 0$$

has only the solution

$$a_1 = a_2 = \dots = a_k = 0$$

If a non-zero solution exists, the vectors are linearly dependent.

Example

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Since

$$v_2 = 2v_1$$

the vectors are linearly dependent.

Matrix Multiplication as Composition

MATRIX MULTIPLICATION AS COMPOSITION

A matrix can represent a linear transformation.

Suppose

$$T_1(x) = Ax$$

and

$$T_2(x) = Bx$$

Applying T_1 followed by T_2 gives

$$T_2(T_1(x)) = B(Ax)$$

Therefore,

$$T_2 \circ T_1 \leftrightarrow BA$$

Matrix multiplication represents composition of transformations.

Example of Composition

Let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

represent horizontal stretching. Let

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

represent vertical stretching. Then

$$\begin{aligned} BA &= \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}} \end{aligned}$$

Thus the combined transformation stretches x by 2 and y by 3.

Matrix Multiplication in Neural Networks

A basic neural-network layer can be written as

$$z = Wx + b$$

where:

- ▶ x = input vector.
- ▶ W = weight matrix.
- ▶ b = bias vector.
- ▶ z = transformed representation.

With an activation function:

$$a = \sigma(Wx + b)$$

Multiple neural-network layers correspond to repeated compositions of transformations.

Three-Dimensional Linear Transformations

THREE-DIMENSIONAL LINEAR TRANSFORMATIONS

A transformation

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

can be represented by a 3×3 matrix.

$$y = Ax$$

For example,

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

scales the x and y directions by 2 while leaving z unchanged.

3D Rotation About the z-Axis

A rotation by angle θ about the z-axis is represented by

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

For $\theta = 90^\circ$:

$$R_z(90^\circ) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Therefore,

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Determinant

DETERMINANT For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

the determinant is

$$\det(A) = ad - bc$$

The determinant provides information about:

- ▶ Scaling of area/volume.
- ▶ Orientation.
- ▶ Invertibility.

For a 2×2 transformation,

$$|\det(A)|$$

gives the factor by which areas are scaled.

Derivation of 2×2 Determinant

Consider

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The determinant is

$$\det(A) = ad - bc$$

If

$$\det(A) = 0$$

the transformation collapses the plane into a lower-dimensional space. Therefore,

$$\det(A) = 0 \Rightarrow A \text{ is singular}$$

and

$$\det(A) \neq 0 \Rightarrow A \text{ is invertible}$$

Determinant and Geometry

For

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

we have

$$\det(A) = 2(3) - 0 = 6$$

Therefore, every area is multiplied by

$$\boxed{6}$$

For example,

$$\text{Original area} = 1$$

becomes

$$\text{New area} = 6$$

This geometric interpretation becomes useful in understanding transformations and Jacobians.

INVERSE MATRICES The inverse of A is a matrix A^{-1} satisfying

$$AA^{-1} = A^{-1}A = I$$

where I is the identity matrix. For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

provided

$$ad - bc \neq 0$$

Derivation of the 2×2 Inverse

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Assume

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Multiplying:

$$\begin{aligned} AA^{-1} &= \frac{1}{ad - bc} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \\ &= \frac{1}{ad - bc} \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} \end{aligned}$$

Therefore,

$$\boxed{AA^{-1} = I}$$

Inverse Matrices in Machine Learning

Linear systems often have the form

$$Ax = b$$

If A is invertible:

$$x = A^{-1}b$$

In ML, matrix inverses appear conceptually in methods such as ordinary least squares:

$$\hat{w} = (X^T X)^{-1} X^T y$$

However, in practical numerical computing, explicitly computing the inverse is usually avoided in favor of solving the linear system.

Column Space

COLUMN SPACE AND NULL SPACE The column space of a matrix is the span of its columns. For

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

the columns are

$$c_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad c_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Since

$$c_2 = 2c_1$$

the column space is

$$\boxed{\text{Col}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}}$$

Null Space

The null space of A is the set of all vectors satisfying

$$\boxed{Ax = 0}$$

For

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

we solve

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

giving

$$x_1 + 2x_2 = 0$$

Thus

$$x_1 = -2x_2$$

Null Space

Let $x_2 = t$:

$$\mathbf{x} = t \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

Therefore,

$$\text{Null}(A) = \text{span} \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}$$

Column Space and Null Space in ML

The column space describes the set of outputs that can be generated by a matrix transformation.

The null space contains directions that are mapped to zero.

In ML, these concepts help understand:

- ▶ Redundant features.
- ▶ Rank deficiency.
- ▶ Underdetermined systems.
- ▶ Identifiability.
- ▶ Linear regression solutions.

A non-trivial null space indicates that different inputs can produce the same output.

Nonsquare Matrices as Transformations

NONSQUARE MATRICES AS TRANSFORMATIONS

A matrix does not need to be square to represent a linear transformation.
For

$$A \in \mathbb{R}^{m \times n}$$

we have

$$A : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

Example:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

is a 2×3 matrix. Therefore,

$$A : \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

Example: Dimension Reduction

Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

and

$$x = \begin{bmatrix} 5 \\ 3 \\ 7 \end{bmatrix}$$

Then

$$Ax = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Example: Dimension Reduction

The transformation maps

$$\mathbb{R}^3 \rightarrow \mathbb{R}^2$$

The third component is discarded. This provides intuition for dimensionality reduction in data processing.

Nonsquare Matrices in AI

Many ML datasets have

n = number of samples

and

d = number of features

so

$$X \in \mathbb{R}^{n \times d}$$

For example:

$$X \in \mathbb{R}^{1000 \times 50}$$

means:

1000 samples, 50 features

Matrix multiplication allows the data to be transformed into different feature dimensions.

DOT PRODUCTS AND DUALITY

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

the dot product is

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3$$

It can also be written as

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$$

Geometric Meaning of Dot Product

The dot product satisfies

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

Therefore:

- ▶ $\theta = 0^\circ$: maximum positive value.
- ▶ $\theta = 90^\circ$: dot product is zero.
- ▶ $\theta = 180^\circ$: negative value.

If

$$\mathbf{u} \cdot \mathbf{v} = 0$$

then

$$\mathbf{u} \perp \mathbf{v}$$

The vectors are orthogonal.

Dot Product in Machine Learning

The dot product is fundamental in ML.

A linear prediction:

$$\hat{y} = \mathbf{w}^T \mathbf{x} + b$$

is a dot product between weights and input features.

Similarity between two vectors can also be measured using the dot product.

For normalized vectors:

$$\|\mathbf{u}\| = \|\mathbf{v}\| = 1$$

we have

$$\mathbf{u}^T \mathbf{v} = \cos \theta$$

This is the basis of cosine similarity used with embeddings.

Duality

A vector can define a linear function through the dot product.

Given

$$w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

define

$$f(x) = w^T x$$

Then

$$f\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = w_1 x_1 + w_2 x_2$$

Thus a vector can be associated with a linear functional.

This relationship between vectors and linear functionals is called **duality**.

CROSS PRODUCTS

The cross product is defined for vectors in $\mathbb{R}^3$.

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}$$

The result is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$.

Geometric Meaning of Cross Product

Magnitude:

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta$$

This equals the area of the parallelogram formed by the vectors.

The direction follows the right-hand rule.

If

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

then

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

CROSS PRODUCTS IN LINEAR TRANSFORMATIONS

Consider a linear transformation

$$x \rightarrow Ax$$

A transformation changes vectors and therefore can change the area of the parallelogram formed by two vectors.

For a 3-dimensional transformation:

$$|\det(A)|$$

gives the volume scaling factor.

Thus determinant and cross product are connected through the geometry of transformations.

Cross Product and Determinant

For vectors u, v ,

$$\|u \times v\|$$

is the area of the parallelogram they define.

If a transformation A is applied to both vectors, the new area scales according to the transformation.

In three dimensions:

$$\det(A)$$

controls oriented volume scaling.

This provides a geometric interpretation of why determinants are important in linear algebra.

Cramer's Rule

CRAMER'S RULE Consider

$$Ax = b$$

where A is a square matrix and

$$\det(A) \neq 0$$

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad b = \begin{bmatrix} e \\ f \end{bmatrix}$$

the system is

$$ax + by = e$$

$$cx + dy = f$$

Cramer's Rule

Cramer's rule gives

$$x = \frac{\begin{vmatrix} e & b \\ f & d \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}$$

$$y = \frac{\begin{vmatrix} a & e \\ c & f \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}$$

Cramer's Rule Example

Solve

$$2x + y = 5$$

$$x - y = 1$$

Coefficient determinant:

$$D = \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -2 - 1 = -3$$

For x :

$$D_x = \begin{vmatrix} 5 & 1 \\ 1 & -1 \end{vmatrix} = -5 - 1 = -6$$

Cramer's Rule Example

Therefore,

$$x = \frac{-6}{-3} = 2$$

For y :

$$D_y = \begin{vmatrix} 2 & 5 \\ 1 & 1 \end{vmatrix} = 2 - 5 = -3$$

Thus

$$y = \frac{-3}{-3} = 1$$

Hence,

$$\boxed{x = 2, \quad y = 1}$$

Change of Basis

CHANGE OF BASIS

A vector can be represented using different bases. Suppose

$$B = [b_1 \quad b_2]$$

is a basis matrix. If

$$[x]_B$$

is the coordinate vector in basis B , then

$$x = B[x]_B$$

Therefore,

$$[x]_B = B^{-1}x$$

provided B is invertible.

Change of Basis Example

Let

$$b_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Then

$$B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Suppose

$$x = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

We need

$$x = ab_1 + bb_2$$

Change of Basis Example

Therefore,

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

giving

$$a + b = 4$$

$$a - b = 2$$

Hence,

$$a = 3, \quad b = 1$$

So

$$[x]_B = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Change of Basis in AI and ML

Different coordinate systems can represent the same underlying data.

This idea is important in:

- ▶ PCA
- ▶ Feature transformations
- ▶ Coordinate transformations
- ▶ Representation learning
- ▶ Computer vision

PCA finds new orthogonal directions in which the data can be represented more efficiently.

These directions are closely connected to eigenvectors of the covariance matrix.

EIGENVECTORS AND EIGENVALUES

A non-zero vector v is an eigenvector of A if

$$Av = \lambda v$$

where λ is the corresponding eigenvalue.

The transformation changes the magnitude of v but does not change its direction.

- ▶ $\lambda > 1$: stretching.
- ▶ $0 < \lambda < 1$: shrinking.
- ▶ $\lambda < 0$: direction reversal.
- ▶ $\lambda = 0$: mapped to zero.

Finding Eigenvalues

Starting with

$$Av = \lambda v$$

we rearrange:

$$Av - \lambda I v = 0$$

Therefore,

$$(A - \lambda I)v = 0$$

For a non-zero v to exist,

$$\boxed{\det(A - \lambda I) = 0}$$

This equation is called the **characteristic equation**.
Its roots are the eigenvalues.

Eigenvalue Example

Consider

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Then

$$A - \lambda I = \begin{bmatrix} 2 - \lambda & 0 \\ 0 & 3 - \lambda \end{bmatrix}$$

Therefore,

$$\det(A - \lambda I) = (2 - \lambda)(3 - \lambda)$$

Set equal to zero:

$$(2 - \lambda)(3 - \lambda) = 0$$

Hence,

$$\boxed{\lambda_1 = 2, \quad \lambda_2 = 3}$$

Finding Eigenvectors

For

$$\lambda_1 = 2$$

solve

$$(A - 2I)v = 0$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Thus

$$v_2 = 0$$

Choose

$$v_1 = 1$$

Finding Eigenvectors

Therefore,

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Similarly, for $\lambda_2 = 3$:

$$v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Eigenvectors in PCA

Principal Component Analysis (PCA) uses eigenvectors of the covariance matrix.
Let

$$S = \frac{1}{n-1} X_c^T X_c$$

Then solve

$$Sv = \lambda v$$

The eigenvectors give principal directions.

The corresponding eigenvalues indicate the amount of variance captured in those directions.

Large eigenvalue $\Rightarrow$ important direction

This is one of the most important applications of eigenvalues in machine learning.

Eigenvectors in Other ML Applications

Eigenvalues and eigenvectors appear in:

- ▶ PCA and dimensionality reduction.
- ▶ Spectral clustering.
- ▶ Graph-based learning.
- ▶ Markov models.
- ▶ Stability analysis.
- ▶ Covariance analysis.

For example, in spectral methods, eigenvectors of a matrix can reveal important structure in data or graphs.

VECTOR SPACES

A vector space is a set of objects together with operations of addition and scalar multiplication satisfying specific axioms.

Examples:

$$\mathbb{R}^2, \quad \mathbb{R}^3, \quad \mathbb{R}^n$$

The set of all n -dimensional real vectors is

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} : x_i \in \mathbb{R} \right\}$$

Subspaces

A subset W of a vector space V is a subspace if:

1. $0 \in W$.
2. If $u, v \in W$, then $u + v \in W$.
3. If $u \in W$ and c is a scalar, then $cu \in W$.

Examples:

- ▶ A line through the origin.
- ▶ A plane through the origin.
- ▶ Column space.
- ▶ Null space.

Dimension and Basis

If a vector space has a basis containing n vectors, then its dimension is

$$\dim(V) = n$$

For example,

$$\mathbb{R}^3$$

has standard basis

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Therefore,

$$\dim(\mathbb{R}^3) = 3$$

Rank and Nullity

For a matrix

$$A \in \mathbb{R}^{m \times n}$$

the rank is the dimension of its column space:

$$\text{rank}(A) = \dim(\text{Col}(A))$$

The nullity is the dimension of its null space:

$$\text{nullity}(A) = \dim(\text{Null}(A))$$

The Rank-Nullity Theorem states:

$$\text{rank}(A) + \text{nullity}(A) = n$$

where n is the number of columns of A .

Rank-Nullity Example

Consider

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

The columns are linearly dependent. Therefore,

$$\text{rank}(A) = 1$$

Since there are 2 columns,

$$\text{rank}(A) + \text{nullity}(A) = 2$$

Hence,

$$1 + \text{nullity}(A) = 2$$

Therefore,

$$\boxed{\text{nullity}(A) = 1}$$

AI/ML APPLICATIONS

Linear Algebra Concept	AI/ML Application
Vectors	Feature representation
Matrices	Datasets and model parameters
Linear combinations	Linear models
Matrix multiplication	Neural-network layers
Dot product	Similarity and prediction
Nonsquare matrices	Dimension mapping
Covariance matrix	Feature relationships
Eigenvectors	PCA
Determinant	Transformation geometry
Null space	Redundancy and identifiability
Basis	Feature representation

Why Linear Algebra Matters in AI

Almost every modern ML model uses linear algebra.

Data $\rightarrow$ Vectors $\rightarrow$ Matrices $\rightarrow$ Transformations $\rightarrow$ Learning

Examples:

- ▶ Neural networks use matrix multiplication.
- ▶ Regression uses vectors and matrices.
- ▶ PCA uses covariance matrices and eigenvectors.
- ▶ Embeddings are represented as high-dimensional vectors.
- ▶ Similarity search uses dot products.
- ▶ Computer vision uses matrix transformations.

Understanding Linear Algebra $\Rightarrow$ Better Understanding of AI/ML

Thank You

Mathematics for Computing

Module 2: Linear Algebra

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